Subject: CSCU
Marks : 64

JECRC UNIVERSITY
BUILD YOUR WORLD

Assignment (Unit-1)

SECTION A (5 questions * 2 marks = 10 Marks)

1. Define phishing and explain two common techniques used in phishing attacks.
2. What is malware? Provide examples of two types of malware and briefly describe their functionalities.
3. Explain the importance of encryption in cybersecurity. Give two examples of encryption algorithms commonly used to secure data.
4. Differentiate between symmetric and asymmetric encryption algorithms. Provide an example of each.
5. Define SQL injection and discuss two potential consequences of a successful SQL injection attack.

SECTION B (3 questions * 7 marks = 21 Marks)

1. Discuss the concept of social engineering in cybersecurity. Provide three examples of social engineering techniques and explain how they can be mitigated.
2. Explain the role of firewalls in network security. Describe two different types of firewalls and compare their functionalities.
3. What is a Distributed Denial of Service (DDoS) attack? Outline three methods used to mitigate the impact of DDoS attacks on a network.

SECTION C (3 questions * 11 marks = 33 Marks)

1. Discuss the steps involved in conducting a vulnerability assessment. Explain how vulnerability scanning tools contribute to this process and provide two examples of such tools.
2. Explain the concept of penetration testing in cybersecurity. Describe the difference between white-box and black-box penetration testing approaches, and discuss the advantages and disadvantages of each.
3. Outline the phases of the Incident Response Process in cybersecurity. Provide a detailed explanation of each phase and discuss the importance of having an effective incident response plan in place.